

William Castle

Software Engineer Contractor

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Professional summary

Full-stack software engineer and recent Computer Science graduate from Virginia Commonwealth University with hands-on experience shipping production web applications end-to-end — from physics-based simulation engines and 3D visualization to authentication, security hardening, and AI-assistant integrations. Comfortable across the stack (TypeScript/React, Node, Python, PostgreSQL, AWS) and drawn to building technically substantial products with real-world impact.

Willing to relocate: Anywhere

Authorized to work in the US for any employer

Work experience

Software Developer

Eiffel Technologies LLC | Remote

September 2025 to Present

Designed and built a full-stack WiFi coverage planning web application that lets network consultants upload building floor plans, visualize physics-based signal heatmaps in 2.5D, and determine optimal wireless access-point placement — then extended it with an AI-assistant-driven planning experience.

- Built a full-stack web application end-to-end (React 19, Three.js/React-Three-Fiber, Zustand, Vite frontend; Node/Express + TypeScript + PostgreSQL backend), delivering ~21,000 lines of production code across v1.0, v1.1, and v2.0 milestones.
- Implemented a physics-based RF signal-propagation engine (ITU-R P.1238 path-loss modeling, material-specific wall attenuation, and multi-band 2.4/5/6 GHz modeling) with real-time heatmap visualization overlaid on a 2.5D building render.
- Developed DXF floor-plan import with layer-based wall selection, scale calibration, and material assignment so consultants can turn CAD drawings into simulation-ready models.
- Engineered interactive access-point placement with drag-to-move APs and live heatmap/coverage recomputation, plus an automatic optimization system (greedy + simulated-annealing) that suggests optimal router positions.
- Offloaded heavy propagation and optimization math to Web Workers, keeping the UI responsive during compute-intensive simulations.
- Built the authentication and API layer (JWT auth, bcrypt password hashing, email verification/password reset, request validation) and completed an OWASP Top 10 security audit and hardening pass.
- Designed and shipped an AI-assistant planning experience: an LLM-style chat (OpenAI, backend-only integration) that visibly calls building-analysis, simulation, and recommendation "toolhooks" to guide users from scenario selection through AP recommendations, with a deterministic fallback and prompt-injection / tool-use boundary safeguards.
- Maintained a test suite across the propagation engine and backend services, and drove production polish including responsive layouts, client branding, and systematic bug-sweep fixes.

AI Incubator Internship

CARE Lab | Richmond, VA

May 2025 to August 2025

Co-facilitated CARELab's "AI for Good" summer incubator at VCU, guiding 5 student teams from initial concept through final pitch as they turned social-impact ideas into practical, AI-enhanced business plans.

- Mentored 5 student teams through the full ideation-to-pitch lifecycle — helping them scope a problem, identify where AI could add genuine value, and shape early ideas into structured, defensible business plans.
- Delivered technical presentations on the AI side of the work, breaking down how the underlying models and approaches functioned so non-technical teams could make informed design decisions.
- Helped define the evaluation criteria used to measure each team's success, giving students concrete benchmarks to build toward and assess their solutions against.

- Coached teams on grounding social-impact missions in realistic, buildable AI solutions rather than hype, and on communicating that value to a non-technical audience.
- Guided teams in preparing and delivering final pitches to a panel of 5 judges drawn from the local business community, with select proposals forwarded to the Mayor's Office.

Education

Computer Science (Undergraduate)

Virginia Commonwealth University | Richmond, VA

August 2022 to May 2026

Skills

GitHub

Python

AWS

Node.js

React

Docker

TypeScript

Next.js

Redis

PostgreSQL